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(54) **MUZZLE BRAKE AND METHOD FOR
MANUFACTURING MUZZLE BRAKE**

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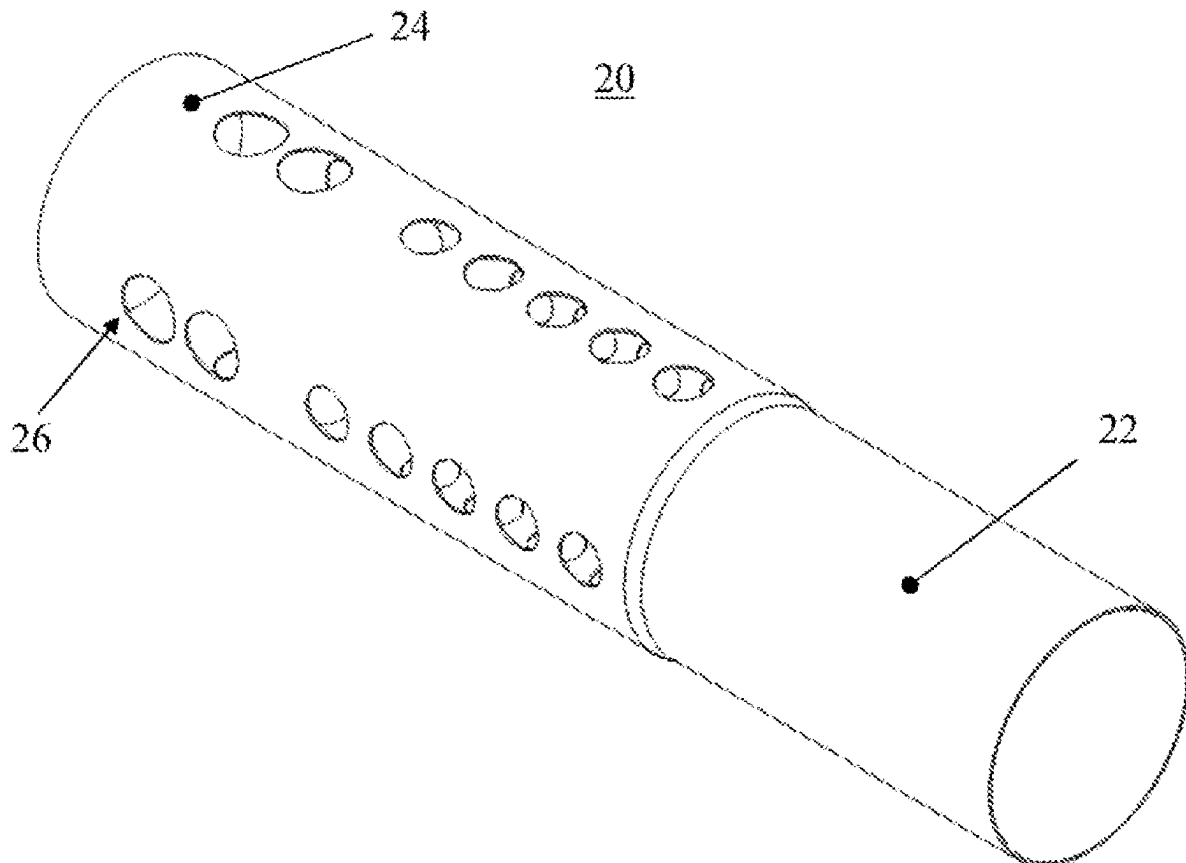
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ABSTRACT

A method for manufacturing a muzzle brake for a barrel includes designing a capsule assembly including a cavity, arranging powder in the cavity in the capsule assembly, pressing the powder so that the powder and capsule assembly are joined together, and removing excess material that covers the opening for outgoing gas flows. A muzzle brake, a firing barrel and a firing device are also provided.



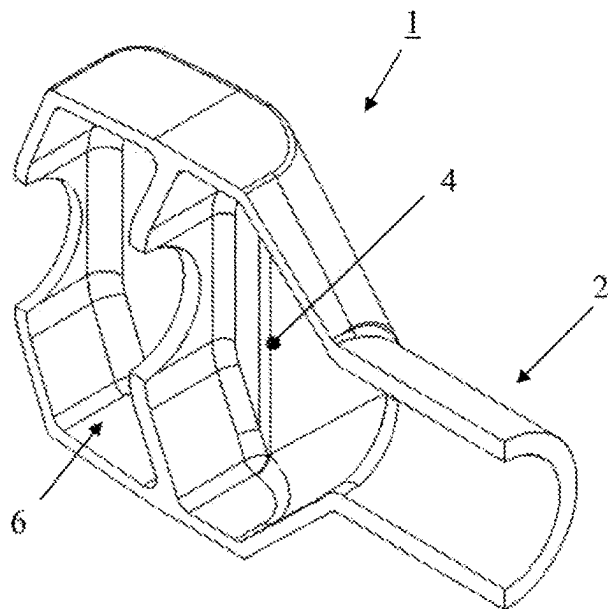


Fig. 1

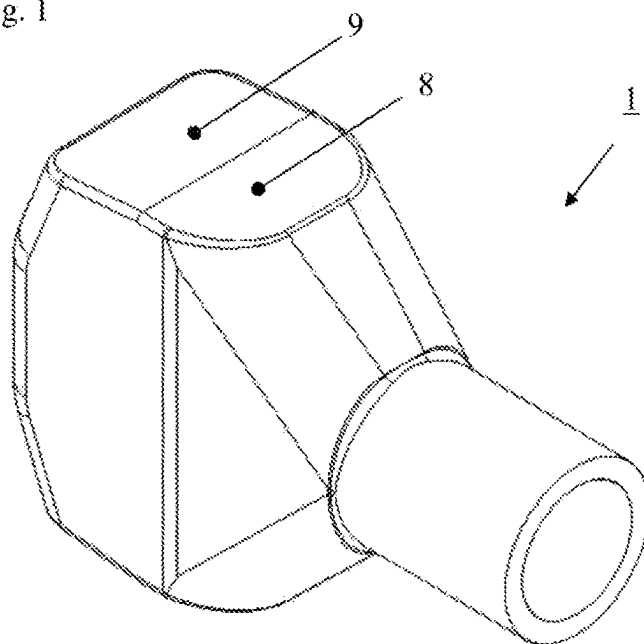


Fig. 2

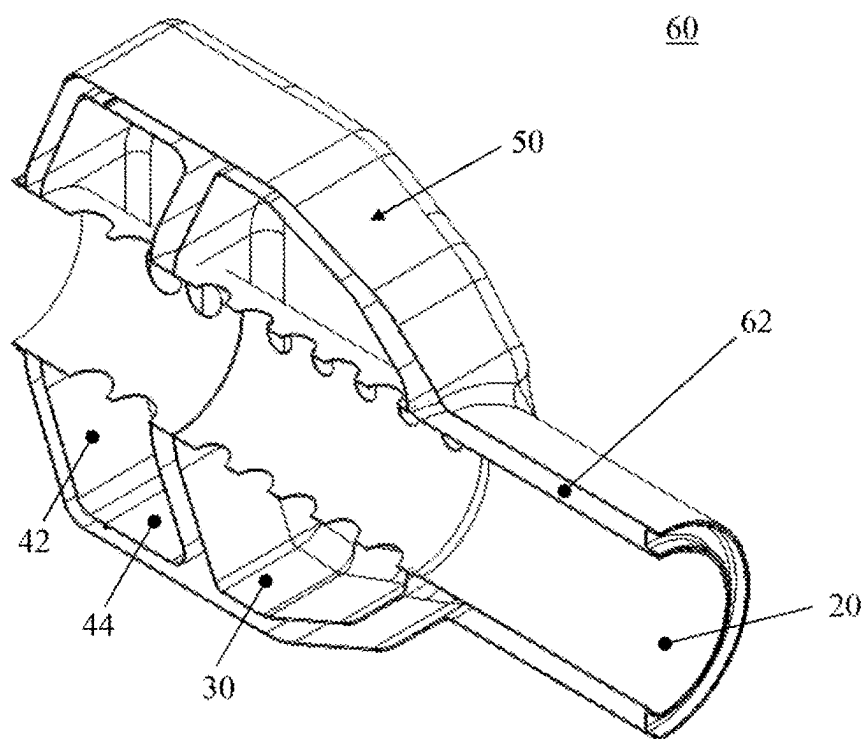


Fig. 3

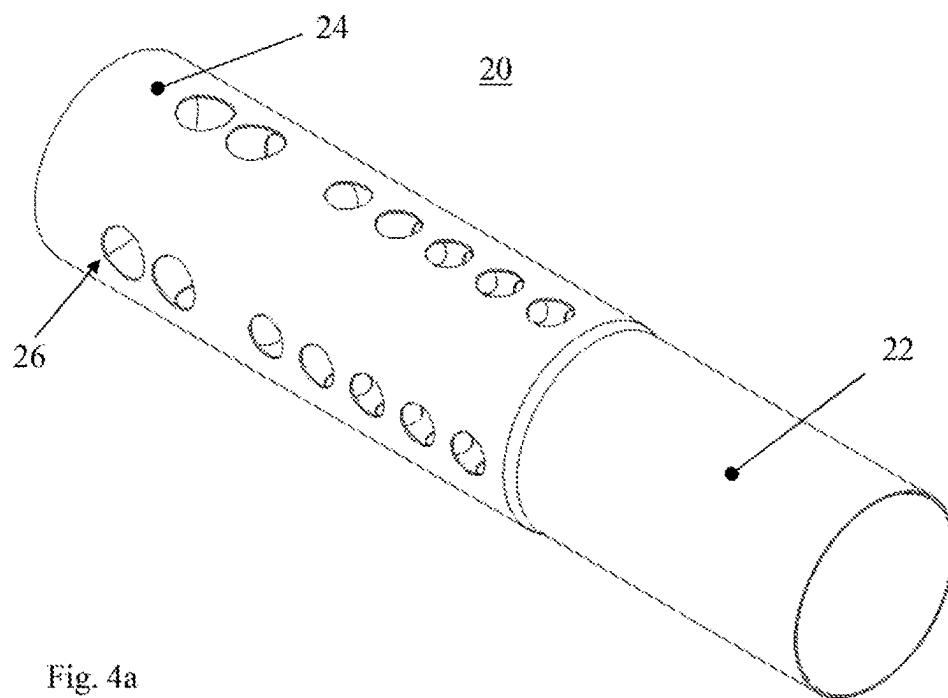


Fig. 4a

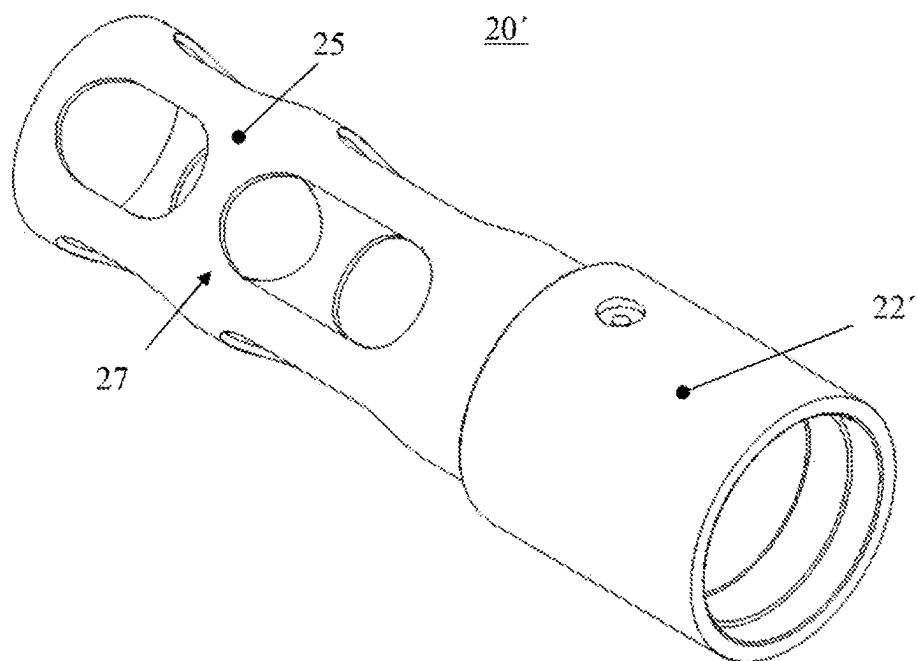


Fig. 4b

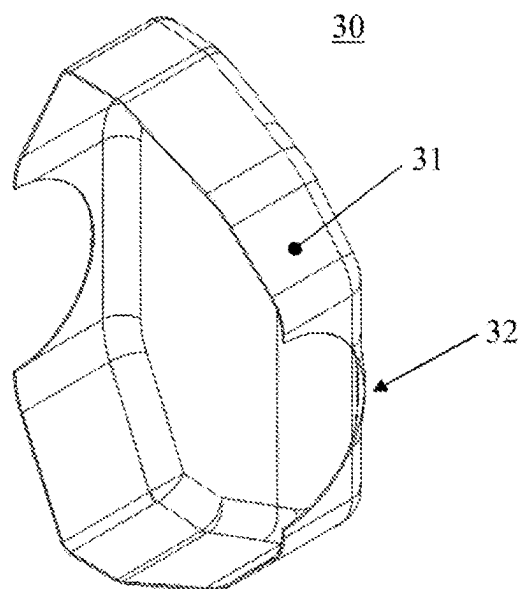


Fig. 5

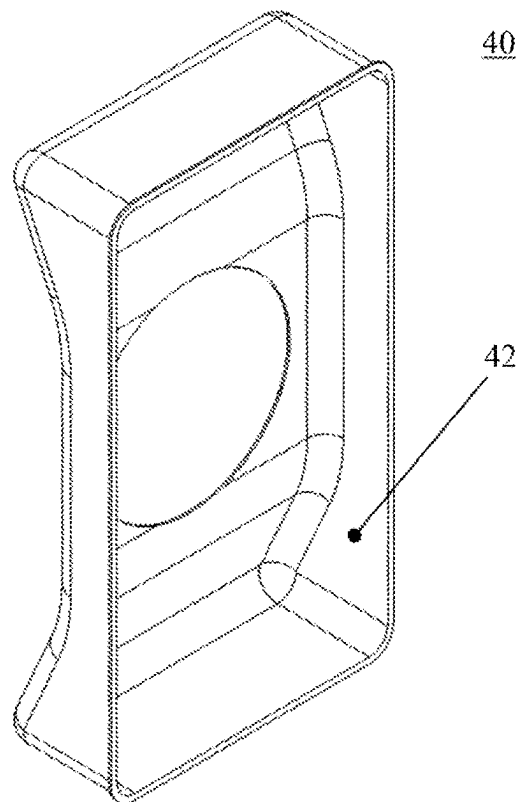


Fig. 6

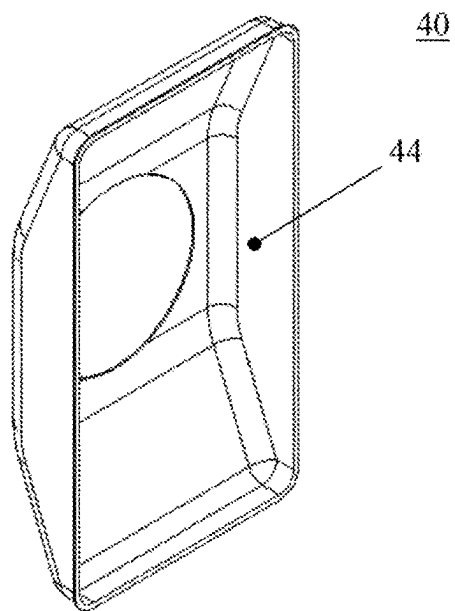


Fig. 7

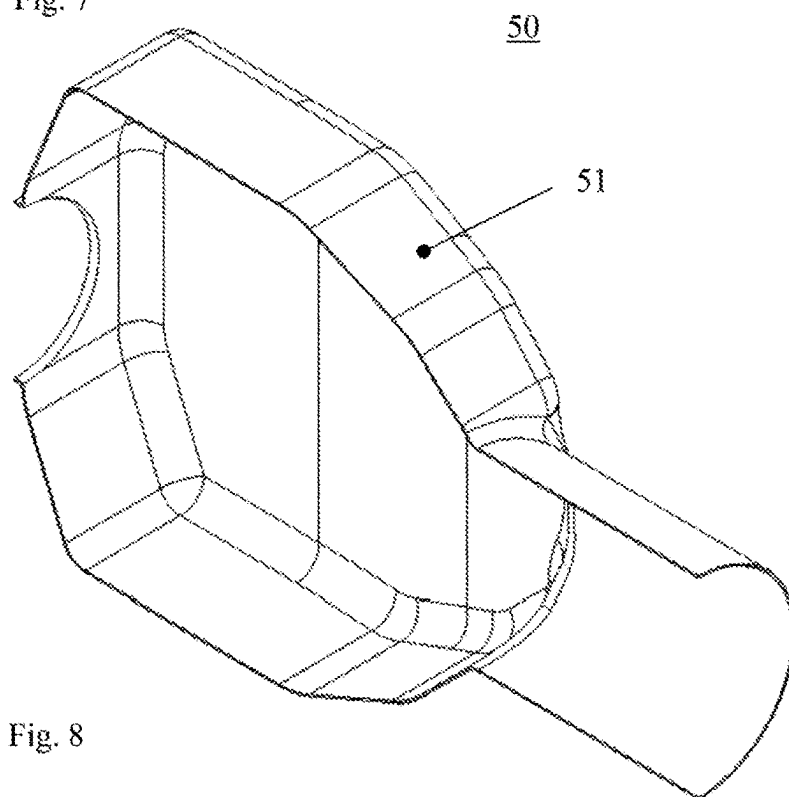


Fig. 8

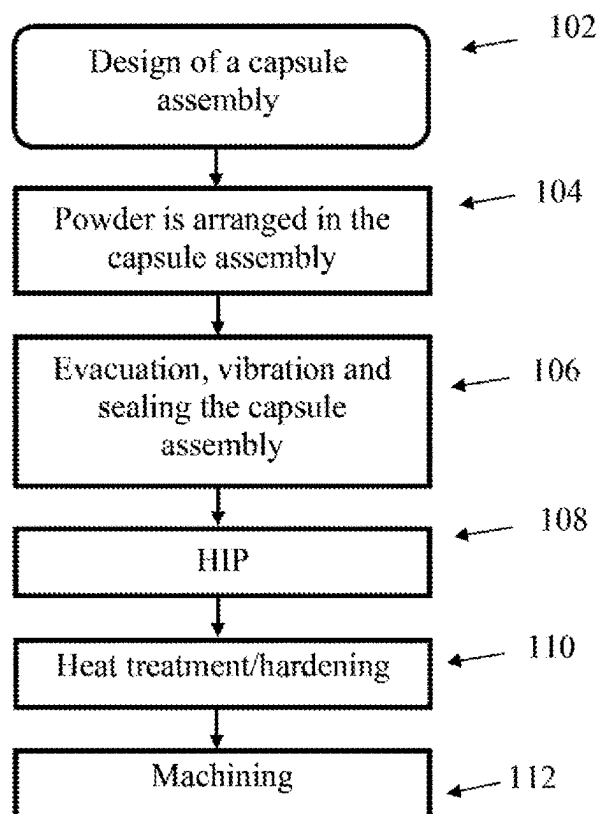
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Fig. 9

MUZZLE BRAKE AND METHOD FOR MANUFACTURING MUZZLE BRAKE

BACKGROUND AND SUMMARY

[0001] The present invention concerns a method for the manufacture of a muzzle brake, a muzzle brake, a firing barrel designed with a muzzle brake, and a firing device designed with a barrel fitted with a muzzle brake.

[0002] When different forms of barrel-based weapons are fired by burning gun powder, which creates a gas expansion and the expanding gas moves a projectile along a barrel, a force is created in the opposite direction of the projectile's direction of motion, commonly known as recoil. Depending on the choice of weapon, gunpowder, projectile, etc., extensive recoil arises under certain conditions, which can be counterbalanced by means of arranging a muzzle brake on the firing barrel. The recoil force also affects the position of the barrel, which is why the barrel's aim may need to be adjusted after a projectile has been discharged. If the recoil can be reduced, and thereby the movement of the barrel reduced, the performance of the weapon system can be improved.

[0003] Muzzle brakes are conventionally manufactured through mill-turning, or by shaping sheet metal components, and placing them on the barrel, primarily by welding. These manufacturing methods are both costly and time-consuming and result in a muzzle brake that is designed based on production limitations and not based on the functional requirements of the muzzle brake. The welding seam between the muzzle brake and the barrel results in poorer mechanical properties than other barrel components, which can result in durability problems. In existing muzzle brake designs, combustion gases also cause extensive erosion to the muzzle brake. This results in high production costs overall.

[0004] An example of a manufacturing method for a muzzle brake is shown in patent document U.S. Pat. No. 9,835,401 B1, which shows a manufacturing method for a muzzle brake that includes creating a casting mold by arranging a ceramic metal in a wax form and then placing liquid metal in the ceramic mold, whereafter a finished muzzle brake can be created.

[0005] The familiar technique above involves extensive manufacturing problems in terms of, e.g., the number of process steps and/or problems related to welding or casting, as well as the amount of material consumption and the design being limited based on manufacturing problems.

[0006] Additional problems which the present invention seeks to solve will become apparent in connection with the following detailed description of the various embodiments.

[0007] It is desirable to achieve a better, simpler, faster and more cost-efficient way of manufacturing a muzzle brake.

[0008] This invention concerns, according to an aspect thereof, a method for manufacturing a muzzle brake for a firing barrel, characterized by the fact that the method comprises the following steps:

- [0009] i.) a capsule assembly is designed including a cavity,
- [0010] ii.) powder is arranged in the cavity of the capsule assembly,
- [0011] iii.) the powder is pressed so that the powder and the capsule assembly are joined together,
- [0012] iv.) excess material that covers the opening for the outgoing gas flows is removed,

[0013] that the powder is pressed using high pressure and heat, also called Hot Isostatic Pressing, HIP,

[0014] that the muzzle brake is heat-treated and hardened using Hot Isostatic Pressing during finishing,

[0015] that the capsule assembly includes an inner tube, a rear capsule, a front capsule, where the inner tube, rear capsule and front capsule are arranged together in an outer capsule,

[0016] that the inner tube is designed with threads,

[0017] that the powder completely or partially comprises refractory metal. Ideally, the powder is arranged so that an outer layer of refractory metal is created on the muzzle brake, which is resistant to erosion. The remainder of the muzzle brake can be manufactured from another powder material, for example, martensitic stainless steel.

[0018] The invention, according to an aspect thereof, also comprises a muzzle brake produced by means of the method described above.

[0019] The invention, according to an aspect thereof, also comprises a firing barrel including a muzzle brake.

[0020] The invention, according to an aspect thereof, also comprises a firing device including a firing barrel including a muzzle brake.

[0021] By manufacturing a muzzle brake with HIP, Hot Isostatic Pressing, a muzzle brake can be achieved with better performance than using previously known technology. Improvements include a homogeneous muzzle brake without weak spots or pores, where none of the components are joined together by welding or casting, for example, and that is designed with a permanent seam against the firing barrel, which results in greater control over performance, fewer process steps and thereby lower manufacturing costs and reduced material consumption because the need for machining each muzzle brake produced is reduced. Furthermore, muzzle brakes can be created with a greater degree of freedom in terms of design.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The invention will be described below by reference to the figures that are included there:

[0023] FIG. 1 shows a cross-section of a HIP blank according to one embodiment of the invention.

[0024] FIG. 2 shows a HIP blank according to one embodiment of the invention.

[0025] FIG. 3 shows a cross-section of the capsule assembly according to one embodiment of the invention.

[0026] FIG. 4a shows the inner tube of the capsule assembly, according to an initial embodiment of the invention.

[0027] FIG. 4b shows the inner tube of the capsule assembly, according to a second embodiment of the invention.

[0028] FIG. 5 shows the inner rear capsule of the capsule assembly, according to one embodiment of the invention.

[0029] FIG. 6 shows the inner front metal plate for the front capsule of the capsule assembly, according to one embodiment of the invention.

[0030] FIG. 7 shows the rear metal plate for the front capsule of the capsule assembly, according to one embodiment of the invention.

[0031] FIG. 8 shows the outer capsule of the capsule assembly, according to one embodiment of the invention.

[0032] FIG. 9 shows the process steps for Hot Isostatic Pressing, HIP, when manufacturing a muzzle brake according to one embodiment of the invention.

DETAILED DESCRIPTION

[0033] The present invention shows an embodiment of a manufacturing method for firing barrels and/or firing barrel components, such as muzzle brakes, using hot isostatic pressing. Hot isostatic pressing, also called HIP, is a production process to control the grain size and structure of the material. HIP also allows metal powder, polymer powder, ceramic powder and composite powder to be pressed into a solid form. The advantages include the fact that all the empty spaces inside metal components that are created through additional manufacturing methods are removed and that mechanical properties such as fatigue resistance/fatigue strength, toughness, plasticity and impact resistance are improved. Furthermore, HIP can create a dense material from metal powder, composite powder, polymer powder or ceramic powder without melting, and materials with partially different characteristics can be combined in the same component.

[0034] Using HIP, a solid material can be created from powder with superior properties because the powder/powder components have a fine, uniform grain size and an isotropic structure. Furthermore, with HIP, different metals can be joined together without needing a temperature-limiting adhesive. Using HIP, several diffusion bonds can be achieved in one process cycle. A large number of metal alloys, as well as polymers and ceramic material could be used in HIP. For example, alloys with nickel, cobalt, tungsten, titanium, molybdenum, aluminum, copper and iron, oxide-and nitride ceramics, glass, intermetallic substances and polymers. HIP enables the bonding and combining of materials that otherwise cannot be combined, i.e. composites.

[0035] An ejection device, also termed a cannon, a howitzer, or a piece, in the sense of an artillery piece, has the goal of making use of a propellant for the purpose of firing a projectile. Preferably, a propellant, such as gunpowder, is initiated in one part of the cannon, oftentimes a chamber specifically adapted to the purpose. Initiation takes place by way of igniting the barrel, for instance by means of an ignition cartridge or an igniter in a munitions device, which is initiated by means of striking. Other methods for igniting the propellant may include ignition of the propellant by means of laser energy or electric energy. The propellant burns at a high rate and results in large amounts of gas being produced, which creates a gas pressure in the chamber which propels the projectile out of the barrel of the firing device. The propellant has been adapted in order to generate a constant pressure on the projectile during the entire barrel procedure, to the greatest extent possible, as the projectile moves in the barrel, which results in the projectile leaving the mouth of the barrel with high speed.

[0036] Projectiles, such as various types of grenades, generally include some form of warhead and some form of barrel which initiates the warhead. Fuzes can be of different types where contact fuzes are common for projectiles that are meant to burst when in contact with an object, timed fuzes when the projectile is meant to burst at a certain predetermined time and proximity fuzes when the projectile is meant to burst when an object comes within a certain distance from the projectile. The use of proximity fuzes is preferred when confronting flying vessels, while timed fuzes can be used when confronting a large number of various different objects. It is advantageous to combine various types of fuze functions in one and the same fuze, for instance

in order for the projectile to burst after a certain time if it fails to detect any object, and so on.

[0037] It is advantageous for the warhead to comprise some type of explosive substance, as well as some type of shattering casing which encloses the explosive substance. Various types of propellants, such as fins, can furthermore be arranged in either the barrel or in its own subcomponent.

[0038] In order to stabilize the projectiles once the projectiles have left the barrel, the projectiles are preferably designed with rotation or with fins. In cases where the projectiles are designed with rotation, the projectiles are said to be rotationally stabilized and in cases where the projectiles are arranged with fins, the projectiles are said to be fin-stabilized. Fin-stabilized projectiles should have no rotation, or low rotation, when leaving the barrel.

[0039] To achieve rotation on the projectiles, the barrel is often designed with rifling, to which the projectile connects during the firing process. Rifling means that the barrel in a firearm, the barrel, is provided with spiral-shaped rifling. The opposite is a smooth-bore barrel. When the rifling engages the projectile during firing, it rotates along its longitudinal axis. Due to the rotation, minor irregularities or damage to the projectile will not cause a drift in the trajectory of the projectile. Rotation is also necessary for an elongated (torpedo-shaped) projectile to maintain its direction after leaving the barrel and not start tumbling around. This is referred to as the projectile being rotation-stabilized. In smooth-bore weapons, only round (spherical) projectiles or fin-stabilized projectiles can be fired. An elongated projectile without fins will tumble as it leaves the muzzle.

[0040] Thus, rifling consists of or comprises grooves that are integrated into the track of the barrel, and the elevation in between is referred to as barriers. The rifling of fine-caliber firearms usually consists of or comprises four grooves that are turned to the right, while cannons, such as artillery pieces, have more grooves depending on the caliber of the launching device. In order for the rifling to be able to engage the projectile, the projectile must either be slightly larger than the diameter between the barriers, which is common for fine-caliber weapons, or be equipped with a special flange, called a belt, which has a slightly larger diameter than the barriers, which is common in projectiles with a diameter greater than 20 mm. The belt can be made out of plastic, composite material or a soft metal, such as brass. The length of the barrel on which the groove rotates an entire revolution is called the pitch and is usually the number of inches per revolution.

[0041] Most barrels include rifling, and, by arranging projectiles with sliding belts, both rotation-stabilized and fin-stabilized projectiles can be launched with rifled barrels. Smooth-bore barrels are basically only used for weapon systems intended to armored combat vehicles, as the rotation of the projectile means that the directed explosive action, RSV, is less effective since the centrifugal force causes the beam to be spread out.

[0042] During firing, as the projectile leaves the barrel, a force arises called recoil, which works in the opposite direction of the movement of the projectile. The recoil affects the firing barrel, and thereby the position of the firing device. The recoil can move the firing barrel, which can result in the barrel needing to be re-aimed after a projectile is discharged. Furthermore, the recoil forces can affect the firing device and cause damage to the firing device over time. By placing a muzzle brake on the firing barrel, part of

the force that is generated from the gunpowder gases that follow the projectile can be used to counteract the recoil force. For example, by directing the gunpowder gases so that they affect the muzzle brake in the opposite direction of the recoil forces. This counteracts the recoil, and a certain level of balance between these forces can be achieved.

[0043] FIG. 1 shows a cross-section of a muzzle brake 1 that is manufactured according to the first embodiment after the Hot Isostatic Pressing stage has been completed.

[0044] The HIP container arranged around the muzzle brake, also called the capsule assembly 60, can also be machined away, for example, via mill-turning or grinding, but can also be retained as a protective sheath for muzzle brake 1. Muzzle brake 1 can now be called a HIP-ed body, or preform, and can be machined to open up the outgoing passage, e.g. for gas flows, and can then be finished into a complete muzzle brake 1 using additional heat treatment, and then can be used as a component on a firing barrel. Muzzle brake 1 can be attached to the barrel using threads on the inside of the connection geometry 2. The muzzle brake in the design shown includes a rear cavity 4 and a front cavity 6, but in alternative embodiments, it can also include one cavity or even several cavities to handle gas flows from the firing barrel.

[0045] FIG. 2 shows muzzle brake 1 in the form of a preform, where the material covering the opening for outgoing gas flow 8 from the first cavity 4 and the opening for outgoing gas flow 9 from the second cavity 6 has not yet been machined off the muzzle brake 1. Machining, such as mill-turning, e.g. cutting or grinding, is preferably performed on preforms, when the material properties are best suited to machining. Furthermore, the connection geometry 2 can be machined, for example, by adding threading. After machining, the preform can be finished into a muzzle brake 1 that is suitable for using on a firing barrel and undergo heat treatment to achieve the correct material properties for a muzzle brake 1.

[0046] FIG. 3 shows a capsule assembly 60, also called a HIP container, in the form of a sheet metal construction, including a cavity 62, where powder can be placed. The powder is arranged freely in the HIP container in the form of the capsule assembly 60. Then with HIP treatment, the powder can be fixed in the intended position in order to create a muzzle brake 1. The capsule assembly 60 in the embodiment displayed consists of or comprises an inner tube 20, a rear capsule 30 and a front capsule 40, consisting of or comprising front plate 42 and a rear plate 44, which are arranged together in the outer capsule 50. Preferably, the rear capsule 30, the front capsule 40 consisting of or comprising a front plate 42 and a rear plate 44, as well as the outer capsule 50, are welded on the inner tube 20 in order to complete the capsule structure 60. The application method is preferably some type of additional manufacturing method where the material can be applied in powder form inside an HIP container in the form of capsule assembly 60, which is a surrounding component designed to hold powder, where powder, as the material to be applied, is arranged freely in the capsule assembly 60. Through continued HIP treatment, the powder can be fixed in the intended position in order to create a muzzle brake 1. Manufacturing methods involving powder have advantages under cramped manufacturing conditions because the material supplied can reach into areas with small dimensions. Capsule assembly 60 is designed with a connecting device, not shown in the diagram, for

evacuating air and vacuum pumping before and/or during the manufacturing process. Capsule assembly 60 is preferably made of some material that experts recognize as suitable for the purpose, usually a metallic material, but it may also be a plastic or a composite, and a variety of materials are already known in the field. In one embodiment, the material in capsule assembly 60 is black sheet plate, and in another embodiment, the material is stainless steel, which also provides a rust-protection function for the muzzle brake 1. The subcomponents of the capsule assembly can be manufactured additionally.

[0047] FIG. 4a shows an inner tube 20 for a capsule assembly 60 in the first embodiment of the invention, where the inner tube is designed as a spacer in the form of a tube. The tube 20 has a connection geometry 22 e.g. that can be designed as threading, for example, to allow threads or blanks for threads to be created on the preform after HIP. Furthermore, tube 20 in the embodiment displayed is designed with a distance geometry 24 that has a larger diameter than the caliber of the firing barrel in order to ensure that projectiles that leave the barrel do not come in contact with the muzzle brake's geometry when the muzzle brake is arranged on the barrel.

[0048] FIG. 4b shows an alternative embodiment of an inner tube 20' for a capsule assembly 60 in the second embodiment of the invention, where the inner tube is designed as a spacer in the form of a tube. The tube 20' is designed with connection geometry 22 which can be designed with threading, as an example. Furthermore, the tube 20' in the embodiment shown is designed with a barrel geometry 25 that has a diameter equal to the caliber of the barrel on which the muzzle brake is placed in order to allow projectiles containing sabots, or other projectiles of a smaller caliber than the barrel, to be fired from the barrel. The barrel geometry 25 is designed with gas outlets 27 in the form of openings in the barrel geometry 25, which allows gas from the firing barrel to stream through the gas outlets 27 and up through the muzzle brake 1. The alternative embodiment of the inner tube 20' is preferably manufactured of material that is the equivalent of the material that the firing barrel is made of and has significantly higher durability requirements and larger dimensions relative to the inner tube 20 according to the first embodiment. The inner tube 20' in the alternative embodiment can be considered an extension of the barrel of the firing device.

[0049] FIG. 5 shows a subsegment 31 of a rear capsule 30 of capsule assembly 60 according to one embodiment of the invention. FIG. 5 shows half of the rear capsule 30 where the penetration hole 32 for the inner tube 20 is visible in the figure. The portion of the figure that is not shown is a symmetrical circle around the penetration hole 32 and mirrors the half of the rear capsule 30 displayed in the figure. The rear capsule 30 is preferably manufactured using two sheet metal components, according to the embodiment shown in FIG. 5, which are joined together into a rear capsule 30. The components for the rear capsule 30 are preferably manufactured using form pressing. A second subsegment 33 is not shown in the figure, but the rear capsule 30 consists of or comprises subsegment 31 and subsegment 33 that are combined into one component.

[0050] FIG. 6 shows a front plate 42 for the front capsule 40 of capsule assembly 60 according to one embodiment of the invention. The front plate 42 is preferably produced by

form pressing sheet metal but can also be manufactured using other methods, e.g. through additional manufacturing. [0051] FIG. 7 shows a rear plate 44 for the front capsule 40 of the capsule assembly 60 according to one embodiment of the invention. The rear plate 42 is preferably produced by form pressing sheet metal but can also be manufactured using other methods, e.g. through additional manufacturing. Front plate 42 and rear plate 44 are joined together into the front capsule 40, e.g. by welding, but can also be joined together using other methods such as gluing, hard-soldering or other joining methods.

[0052] FIG. 8 shows a first subsegment 51 of an outer capsule 50 of the capsule assembly 60 according to one embodiment of the invention. FIG. 8 shows half of the outer capsule 50. The part of the figure that is not shown, the other subsegment 52, mirrors the half of the outer capsule 50 displayed in the figure. The rear capsule 50 is preferably manufactured using two sheet metal components, subsegment 51 and subsegment 52, according to the embodiment shown in FIG. 8, which are joined together into an outer capsule 50. The components for the outer capsule 50 are preferably manufactured using form pressing.

[0053] The capsule assembly 60 is therefore finished by arranging the inner tube 20, 20', rear capsule 30 and front capsule 40 in the outer capsule 50. Preferably, the rear capsule 30, consisting of or comprising subsegment 31 and subsegment 33, the front capsule 40 consisting of or comprising a front plate 42 and a rear plate 44, as well as the outer capsule 50, consisting of or comprising subsegment 51 and subsegment 52, are welded on the inner tube 20, 20' in order to complete the capsule structure 60.

[0054] FIG. 9 shows the manufacturing method 100 for muzzle brake 1. The inner tube 20, the rear capsule 30, as well as the front capsule 40 are arranged in the outer capsule 50 to create a capsule assembly 60 in the Formation stage of a capsule assembly 102. A capsule assembly 60, also called a HIP container, is a device in which powder is arranged in a manner that allows it to be shaped into a HIPed body under high temperature and high pressure. Powder in the capsule assembly 60 is placed in the capsule assembly 104 during the Powder stage. After the powder material has been placed in the capsule assembly 60, the capsule assembly 60 is evacuated, vibrated and sealed in order to evenly distribute the powder in the capsule assembly 60 during the Evacuation, Vibration and Sealing step for capsule assembly 106. HIP is then carried out in the HIP 108 stage, i.e. a gas is used to create isostatic pressure in the capsule assembly 60 via a connecting device on capsule assembly 60 that supplies the gas. Before the gas is supplied to the capsule assembly, the capsule assembly is vacuum-pumped or otherwise evacuated of air or the filling gas/fluid placed in the capsule assembly 60 prior to evacuation, e.g. by rinsing the assembly with a noble gas. The entire capsule assembly 60 is simultaneously heated to create a preform or a HIPed body. The HIP temperature is preferably 20% below the melting temperature for the material; for martensitic steel, the HIP temperature is approximately 1100° C., which is 80% of the material's melting point. The isostatic pressure preferably exceeds 150 MPa. After the hot isostatic pressing is completed, the body can undergo heat treatment/hardening 110, which means that the now merged body is heated. After heat treatment, the material is suitable for machining, e.g. mill-

turning, as well as threading the connection geometry for the muzzle brake and machining away excess material, e.g. material that covers the opening for the outgoing gas flows and any other parts of the HIP container, in the Machining step 112.

[0055] After the muzzle brake 1 is finished, the muzzle brake can be attached to a firing barrel, e.g. using the threading in the muzzle brake. Other ways of attaching the muzzle brake to the barrel can be, but are not limited to, welding, shrink-wrapping or other attachment methods.

[0056] A finished muzzle brake must achieve an impact strength exceeding 27 J at -40° C., the tension yield point should exceed 650 MPa and fall within the interval of 650 MPa to 1200 MPa in one embodiment.

[0057] Furthermore, a finished muzzle brake must withstand a pressure level of up to 70 MPa and a temperature of approx. 2000 K.

[0058] The material used for the powder is preferably martensitic stainless steel with high concentrations of chromium and nickel, potentially with refractory material forming a layer on the muzzle brake in order to better withstand erosion from gun powder gases.

[0059] The invention is not limited to the embodiments specifically shown, but can be varied in different ways within the framework of the claims.

[0060] For instance, it is clear that the choice of material, choice of geometric forms, the elements and details included in the muzzle brake, are adapted to the weapons system(s), platforms and other construction-related properties that are applicable at this time.

[0061] Furthermore, all types of firing barrels, including small caliber, medium caliber and large caliber barrels are included.

1. A method of manufacturing a muzzle brake for a barrel, comprising
 - providing a capsule assembly including a cavity,
 - arranging powder in the cavity of the capsule assembly (60),
 - pressing the powder so that the powder and the capsule assembly are joined together,
 - removing excess material covering an opening for outgoing gas flows.
2. Method according to claim 1, wherein the powder is pressed using high pressure and heat.
3. Method according to claim 2, wherein the muzzle brake is heat treated and hardened using Hot Isostatic Pressing when finishing a muzzle brake.
4. Method according to claim 1 wherein the capsule assembly includes an inner tube, a rear capsule, and a front capsule, where the inner tube, the rear capsule, and the front capsule are arranged together in an outer capsule.
5. Method according to claim 4, wherein the inner tube includes threads.
6. Method according to claim 1 wherein the powder comprises refractory metal.
7. A muzzle brake produced by a method specified in claim 1.
8. A firing barrel comprising a muzzle brake according to claim 7.
9. A firing device comprising with a barrel according to claim 8.

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